

FCC Unit Gasoline Yield Analysis & Optimization

Diagnosing and Optimizing Gasoline Yield in a Fluid Catalytic Cracking Unit Using Python, SQL, and Power BI

Problem Statement

The Fluid Catalytic Cracking (FCC) unit is the most economically critical conversion unit in a petroleum refinery. Gasoline yield losses occur when key process variables, particularly the Riser Outlet Temperature (ROT) and catalyst-to-oil ratio drift outside their optimal ranges. This project investigates which variables drive yield variability, diagnoses the specific operating conditions associated with poor performance, and defines an operating window that maximizes gasoline yield while maintaining an acceptable balance of LPG, diesel, and coke production.

Aim

To identify the process variables driving gasoline yield inefficiency in an FCC unit and determine the optimal operating conditions that maximize gasoline yield while maintaining an acceptable LPG, diesel, and coke balance.

Tools Used

Python SQL Power BI Excel

Headline Results

3.72 ppts Performance Gap (High vs Low Groups)	46.80% Gasoline Yield in Optimal Window	+3.23 ppts Yield Improvement vs Worst Case
535–550°C Optimal ROT Range	5.5–6.0 Optimal Cat-to-Oil Ratio	+1.60 ppts Yield Improvement vs Baseline

Key Findings

Diagnosis

- Riser Outlet Temperature (ROT) is the dominant driver of gasoline yield (Pearson $r = +0.58$)
- Catalyst-to-oil ratio above 7.0 causes secondary cracking into coke, reducing gasoline yield to 43.85%
- Operating below 525°C ROT and above 6.5 cat-to-oil simultaneously produces the dataset worst-case yield of 43.57%
- A performance gap of 3.72 percentage points separates the High and Low yield operating groups

Optimization

- Optimal window: ROT 535–550°C and catalyst-to-oil ratio 5.5–6.0
- Average gasoline yield inside the optimal window: 46.80%
- LPG yield also improves (+0.73 ppts) inside the optimal window
- Coke yield reduces from 7.03% outside to 6.54% inside - heat balance maintained
- Slurry yield reduces from 4.46% to 3.85% - better overall conversion confirmed

Key Visualizations

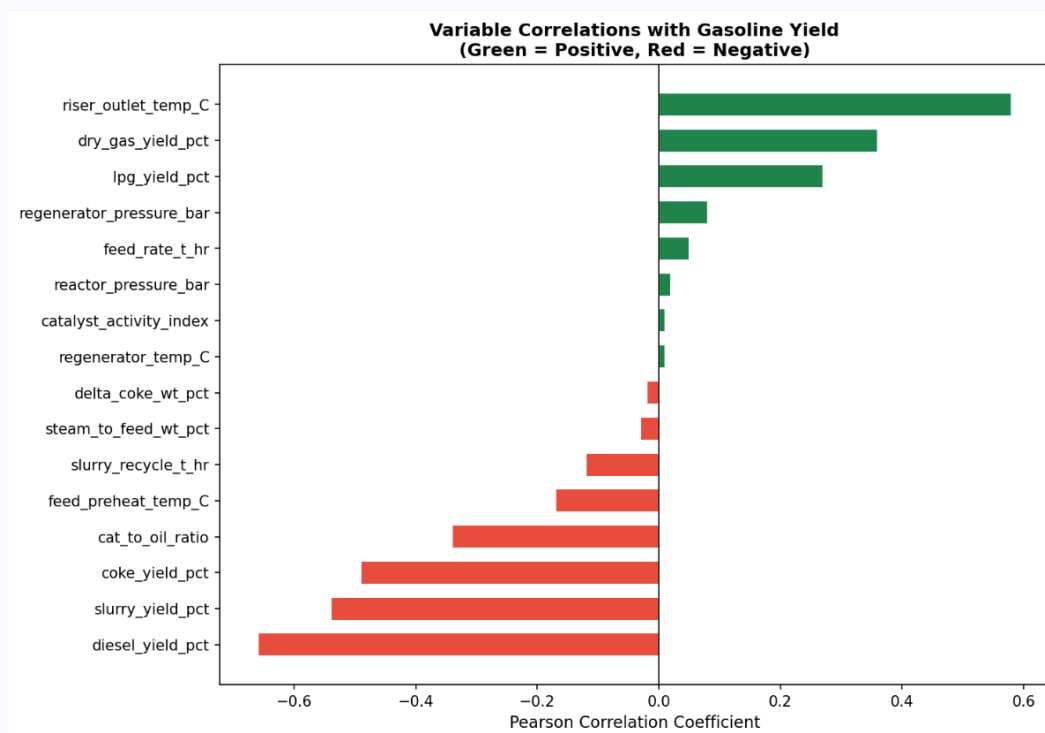


Figure 10: Ranked correlation chart — variable correlations with gasoline yield

Pearson correlation coefficients for all variables against gasoline yield. ROT ($r = +0.58$) is the strongest positive driver; diesel yield ($r = -0.66$) is the strongest negative relationship.

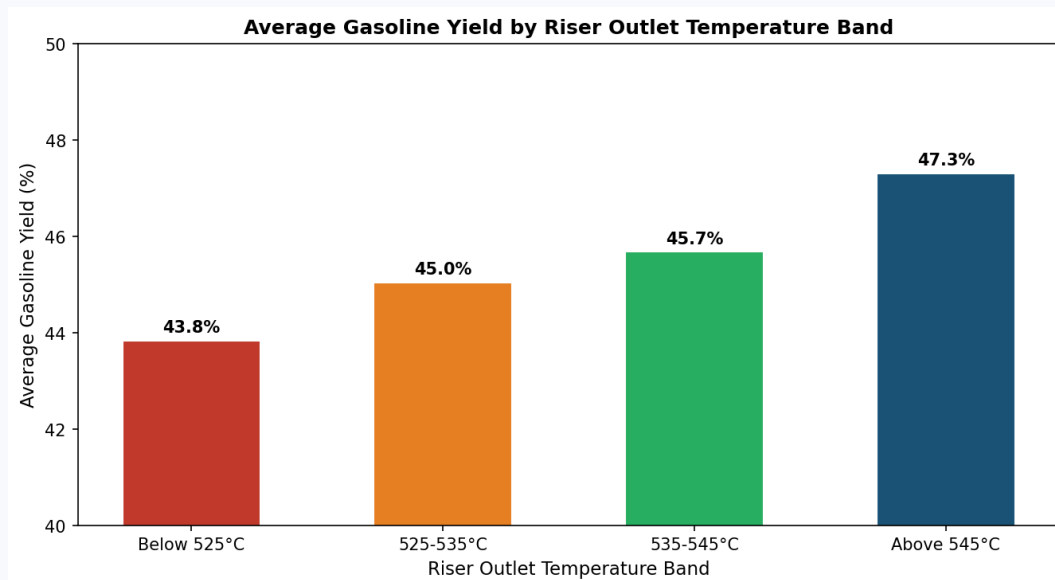


Figure 12: Average gasoline yield by ROT band

Gasoline yield rises monotonically from 43.5% (Below 520°C) to 47.3% (Above 545°C) across seven ROT bands. Red = underperforming, yellow = transitional, green = optimal.

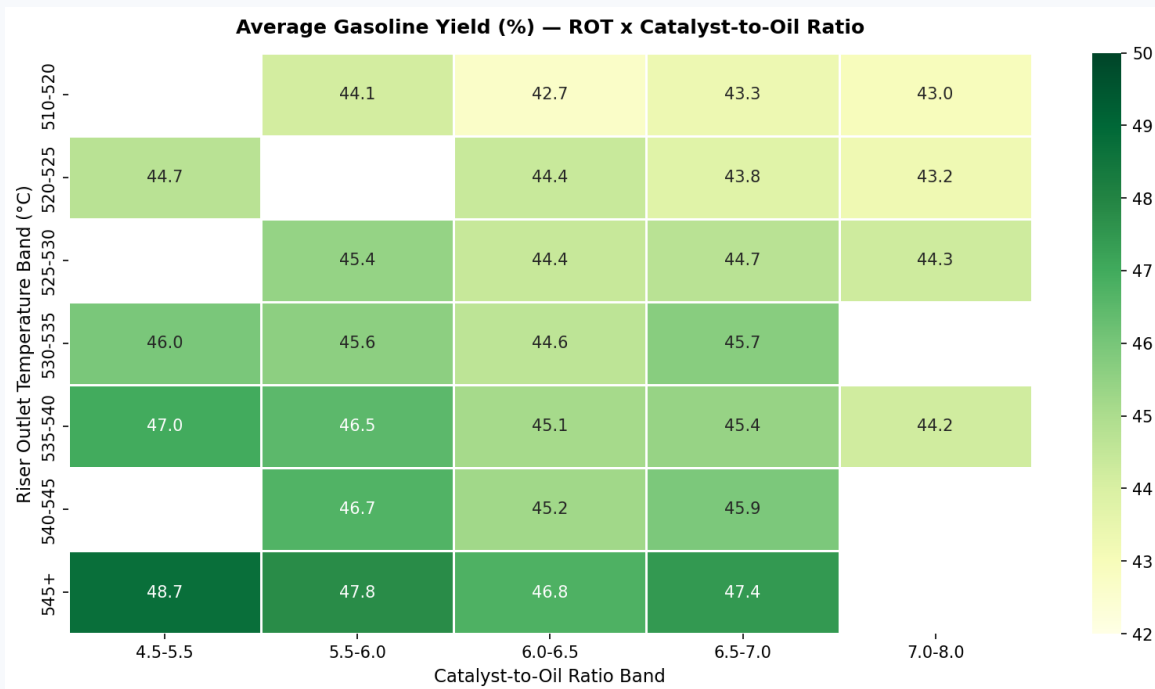


Figure 16: ROT x Catalyst-to-Oil Ratio heatmap

Average gasoline yield across all ROT and catalyst-to-oil ratio combinations. Darkest green cells (bottom-left) represent the optimal operating zone.

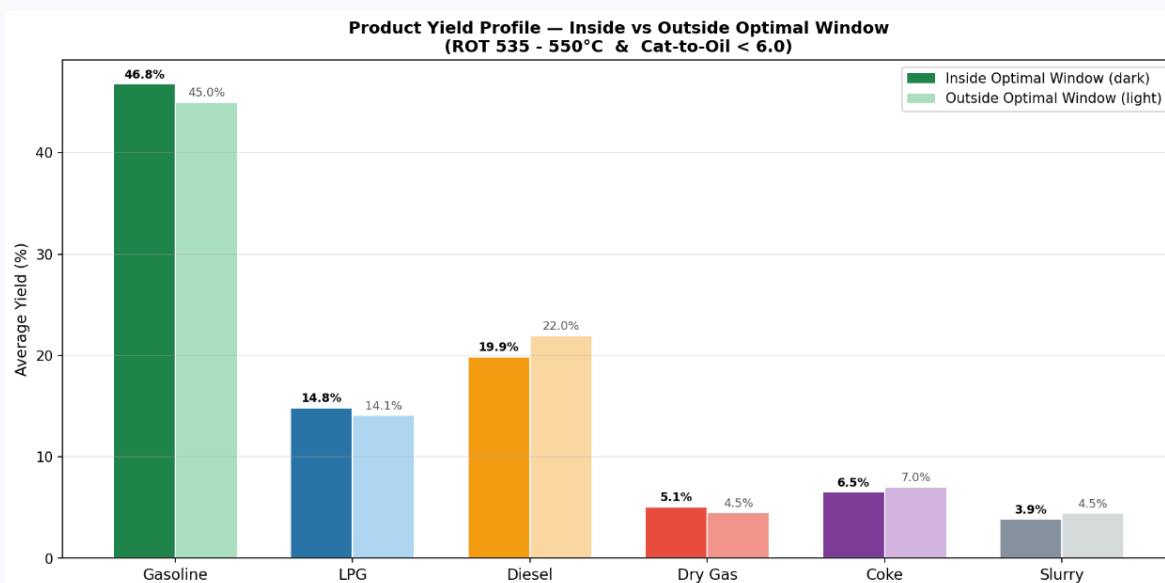


Figure 15: Product yield profile — Inside vs Outside optimal window

All desirable yields (gasoline, LPG) are higher inside the optimal window; all undesirable yields (diesel, coke, slurry) are lower or stable. Dry gas marginally higher but within acceptable limits.

Operational Recommendations

- Maintain ROT within 535–550°C to maximize gasoline yield
- Maintain catalyst-to-oil ratio within 5.5–6.0; avoid sustained operation above 7.0
- Treat ROT below 528°C combined with cat-to-oil above 6.5 as a priority alarm condition
- Monitor dry gas yield as a leading indicator of over-cracking when ROT exceeds 545°C
- Accept a diesel yield reduction of 2–4 percentage points as an expected trade-off in the optimal gasoline window

Methodology Summary

This project followed a structured five-phase analytical workflow, applying a combination of Excel, Python, SQL, and Power BI across each phase.

Phase 1 — Data Collection & Cleaning

The dataset was generated using operating parameters and variable ranges from Zhang et al. (2021). The raw dataset comprised 106 rows and 20 columns. Data quality issues identified and resolved included: string sensor errors in 4 numeric columns, 2 unit inconsistency errors in ROT (°F converted to °C), 7 physically implausible outlier values, 38 missing values across 8 columns, 6 duplicate rows, and inconsistent date formats. The final clean dataset contained 95 rows and 18 columns with zero missing values and zero duplicates.

Phase 2 — Exploratory Data Analysis (EDA)

Outlier detection using IQR-based boxplots confirmed the cleaning results. Univariate analysis (histograms) characterized the distribution of all 11 process variables and the target variable (gasoline yield). Bivariate scatter plots examined the relationship between each process variable and gasoline yield. A Pearson correlation heatmap and ranked bar chart identified ROT as the strongest positive driver ($r = +0.58$) and diesel yield as the strongest negative relationship ($r = -0.66$).

Phase 3 — Performance Diagnosis

The dataset was segmented into High ($\geq 45.54\%$), Medium, and Low ($\leq 44.57\%$) yield groups using percentile thresholds. Five SQL queries were executed using pandasql to compare mean operating conditions between groups, segment yields by ROT and catalyst-to-oil bands, and isolate the worst-case combined operating profile. Key finding: a 3.72 percentage point performance gap separates High and Low yield groups, driven primarily by differences in ROT and catalyst-to-oil ratio.

Phase 4 — Optimization Analysis

Fine-grained segmentation at 5°C ROT intervals and 0.5-unit cat-to-oil intervals narrowed the optimal operating ranges. The combined optimal window (ROT 535–550°C, cat-to-oil 5.5–6.0) was validated against the full dataset, confirming a 46.80% average gasoline yield and acceptable co-product balance. The yield improvement over worst-case conditions was quantified at +3.23 percentage points.

Phase 5 — Dashboard Development

A three-page interactive Power BI dashboard was built and connected to the cleaned dataset. Page 1 (Performance Overview) presents KPI cards, yield by performance group, and an ROT vs yield scatter plot. Page 2 (Diagnosis) presents the operating condition comparison table, ROT band chart, and catalyst-to-oil band chart. Page 3 (Optimization) presents the optimal window specification, before/after yield comparison, product yield profile, and operational recommendations.

Reference

Zhang, Y., Li, Z., Wang, Z., & Jin, Q. (2021). Optimization study on increasing yield and capacity of fluid catalytic cracking (FCC) units. *Processes*, 9(9), 1497. <https://doi.org/10.3390/pr9091497>

Full research report, Python notebook, SQL queries, and dataset available on request.